

Effect of four training systems on the temporal dynamics of downy mildew in two grapevine cultivars in southern Brazil

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Abstract The aim of this study was to evaluate the temporal dynamics of downy mildew epidemics under four training systems: vertical shoot positioning (VSP), Geneva Double Curtain (GDC), Simple Curtain (SC) and Tendone in two cultivars ('Cabernet Sauvignon' and 'Merlot'). Experiments were conducted at commercial vineyards in southern Brazil, during the 2012–2013 and 2013–2014 growing seasons. Downy mildew incidence and severity were quantified on a weekly basis from the first symptoms appearance until harvest. Training systems and cultivars were compared based on the following variables of downy mildew progress curves: a) beginning of symptoms appearance (BSA); b) maximum disease incidence and severity ($I_{\max}$, $S_{\max}$); c) time to reach maximum disease incidence/severity (TRMDI and TRMDS); and d) area under incidence and severity disease progress curve (AUIDPC and AUDPC). The training systems significantly affected AUDPC and $S_{\max}$. AUDPC was lowest in the VSP system in all cultivars and growing seasons. In addition, 'Merlot' showed significantly lower AUIDPC and AUDPC than 'Cabernet Sauvignon' in both growing seasons. The Tendone and GDC systems showed significantly higher AUDPC in all cultivars and growing seasons. Our results suggested that the use of the VSP system contributes to reduce

downy mildew severity on 'Merlot' and 'Cabernet Sauvignon' in southern Brazil.

Keywords *Vitis vinifera* · *Plasmopara viticola* · Merlot · Cabernet Sauvignon · Disease progress curve · Cultural control

Introduction

Grape (*Vitis vinifera* L.) canopy architecture and microclimate can be affected by both training system and cultivar (Pugliese et al. 2011; Calonnec et al. 2013). Downy mildew [*Plasmopara viticola* (Berk. & Curt) Berl. & de Toni] is among the most devastating disease of grapevine in the world, especially in southern Brazil, with potential crop losses up to 100 % (Chavarria et al. 2007). Some combinations of training system and cultivar can help to reduce progress of downy mildew, especially in regions where weather conditions are usually favorable for the disease such as in the highlands of Santa Catarina State in southern Brazil.

Downy mildew affects both the quantity of harvest and the quality of wine produced from infected grapes. The pathogen's life cycle is strongly influenced by weather conditions. It overwinters as oospores in fallen grape leaves and produces zoosporangia (conidia) in spring. In the presence of water, zoosporangia eject zoospores that travel to leaves or bunches by rain splashes. Zoospores can cause primary infections when the host surface remains wet for a sufficient time period. Dry periods are able to kill zoosporangia and zoospores. Also, the secondary infection cycles, caused by zoosporangia produced on affected host tissue, need wetness to cause infection. The minimum temperature for pathogen growth is 9–10 °C, while temperatures higher than 34–35 °C inhibit the pathogen (Kennelly et al. 2005; Austin and Wilcox 2011; Boso et al.

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2011). Losses can range from 40 to 90 % depending on the inoculum level, weather, and cultivar susceptibility. Cultural and genetic control measures are important components of integrated pest management (IPM) and may help to reduce fungicide applications or use of more environmentally safe fungicides (Yu et al. 2012).

Although the use of IPM is desirable, disease control of downy mildew is generally achieved by fungicide use, but it is not always possible or effective owing to the phenological stage of the plant, preharvest spray interval restrictions, and/or environmental conditions such as high annual rainfall. Alternatives to chemicals, such as training systems and pruning together with some cultivar tolerance have been shown to minimize leaf wetness, susceptibility and hence reduce downy mildew severity (Howel et al. 1991; Alonso-Villaverde et al. 2011; Yu et al. 2012). Additional cultural practices include basal leaf removal, shoot positioning, trellising and use of cultivars with different plant architecture, which have been shown to improve production by altering canopy microclimate (Zahavi et al. 2001; Pugliese et al. 2011). Benefits of using these cultural practices include increased exposure of the foliage to light, improved air movement, and potential for mechanization of pruning and harvesting (Dokoozlian and Kliever 1995; Kennelly et al. 2005). Many of these practices are part of canopy management, a term that refers to how the architecture of the leaf canopy may be manipulated in order to beneficially alter the microclimate around the vine (Howel et al. 1991; Giorio and Nuzzo 2012).

The advancements in vineyard design, trellis and training systems, and canopy management practices have dramatically improved wine grape productivity and fruit quality all over the world (Austin and Wilcox 2011; Calonnec et al. 2013). There are four canopy training systems employed in the highlands region of southern Brazil: vertical shoot positioning (VSP) trellis, Geneva double curtain (GDC), single curtain (SC), and Tendone (Miele et al. 2009). Each training system may have a different effect on different cultivars, due to phenotypic traits of each cultivar. For example, ‘Merlot’ and ‘Cabernet Sauvignon’ are the two cultivars commonly grown in the highlands of southern Brazil and both are susceptible to downy mildew. Merlot vine has medium-high vigor and a trailing growth habit. With a long vegetative growth cycle, it tends to produce dense, shaded canopies due to lateral shoot development. Consequently, it is susceptible to several bunch rot and reduced winter cold tolerance (Reynolds et al. 2004; Miele et al. 2009; Pugliese et al. 2011). In comparison, ‘Cabernet Sauvignon’ is perhaps the most vigorous cultivar commercially grown in this region, often resulting in excessive vegetative growth. This high vigor, if not managed, can cause canopy shading and related problems of diseases and fruit quality (Zahavi et al. 2001; Yu et al. 2012).

The choice of an effective combination of cultivar and training system that contributes to reduce disease severity

and consequent losses is needed (Reynolds et al. 2004; Austin and Wilcox 2011; Calonnec et al. 2013). The effect of different training systems combined with different cultivars on downy mildew development has not been well examined and there are few reports in the literature, especially in the highlands of southern Brazil. Thus, the aim of this study was to quantify the effects of training systems (VSP, GDC, SC, and Tendone) on downy mildew disease development in the two grape cultivars, ‘Merlot’ and ‘Cabernet Sauvignon’, grown in the highlands region of southern Brazil, during the growing seasons of 2012–2013 and 2013–2014.

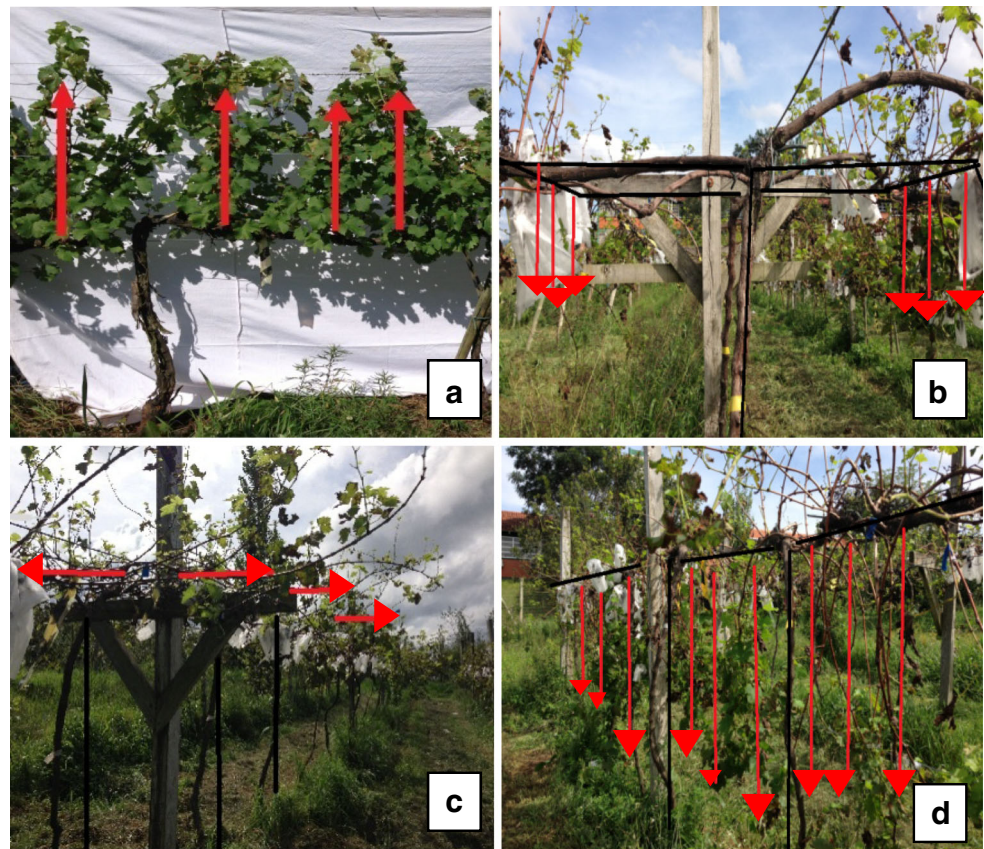
Material and methods

Experiments were carried out in two nearby (~6 m) vineyards located in Lages municipality, Santa Catarina (SC), southern Brazil, during the 2012–2013 and 2013–2014 growing seasons. The vineyards are located at 27°48'39" S by 50°19'W, 916 m elevation. The climate is humid mesothermic according to the Köppen classification (Peel et al. 2007) and soil type is cambisol, which is characterized as having high clay (432 g kg⁻¹) and organic matter (67 g kg⁻¹). Daily rainfall, relative humidity, and hourly temperatures were recorded at the Santa Catarina Hydrology and Environmental Resources Center – Epagri (Epagri 2015). The weather station (Davis Pro2-6153- Onset Computer Corp., Pocasset, MA, USA) was at 500 m distant from the vineyards.

There were approximately 450 grape plants per vineyard in 60–70 rows of 8-year-old *V. vinifera* cvs. ‘Cabernet Sauvignon’ and ‘Merlot’ grafted on rootstock cv. ‘Paulsen 1103’. Both cultivars are susceptible to downy mildew (Lalancette et al. 1988; Madden et al. 2000; Kennelly et al. 2005; Miele et al. 2009). The plant density was approximately of 2.7 plants/ha. Each row consisted of approximately 30 individual vines. There were approximately 15 rows of each cultivar in each trellis system (total of 60 rows per cultivars) that were trained to either a vertical shoot positioning trellis (VSP), Geneva double curtain (GDC), simple curtain (SC) or Tendone at distances of 3.0 × 1.2 m (Fig. 1). The trellis systems were replicated within each variety block. A completely randomized experimental block consisted of five vines along of the rows. These were replicated three times for each treatment. Each treatment was separated by buffer rows and used in each of two consecutive 2012–2013 and 2013–2014 growing seasons. Three vines at the end each plot were necessarily excluded, as were vines at the end of each row, and all vines in border rows.

The VSP and SC training systems were pruned by cordon spur with 0.80 and 1.2 m cordon height above ground, respectively and 1.0 and 1.6 m of height from the ground to the fruiting zone, respectively (Fig. 1). The GDC and Tendone were mixed pruned (2–3 buds/short spur, 15 spurs per vine

Fig. 1 Different training systems with indication of annual branches training for: Vertical Shoot Positioning –VSP (a), Geneva Double Curtain – GDC (b), Tendone (c), and simple curtain (d) used during the 2012/13 and 2013/14 growing seasons experiments



and 7–8 buds/long spur, 21 spurs per vine) within a vine. The GDC and Tendone were 1.3 m and 1.8 m cordon height above ground, respectively and 1.3 and 1.6 m of height from the ground to the fruiting zone, respectively (Fig. 1). The date of pruning occurred from July to August according to training systems.

The fungicides captan/CAPTAN 500WP (dicarboximide, 500 mL of active ingredient (a.i)/L⁻¹, 240 g of commercial product (c.p)/ha; mancozeb/DITHANE NT (dithiocarbamate, 800 mL a.i/L⁻¹, 350 g/c.p/ha); fenamidone/CENSOR SC (imidazole, 500 mL i.a/L⁻¹, 30 mL/c.p/ha), and thiophanate methyl/CERCOBIN 700WP (benzimidazole, 700 mL i.a/L⁻¹, 70 mL/c.p/ha) were applied in rotation by spray from September to March (beginning of bud burst to harvest) at a frequency of 10 and 12 times in the growing seasons of 2012–2013 and 2013–2014, respectively, to maintain low levels of downy mildew and other foliar diseases, yet allowing sufficient disease to evaluate training systems and cultivars. Vineyards were irrigated by trickle and maintained as recommended to commercial growers in this region (20 L/day/plant) (BRASIL 2013).

Each plot was assessed weekly for incidence and severity of downy mildew beginning at first symptom appearance (end of sprouting/October) and continuing until harvest (February/March) during the 2012–2013 and 2013–2014 growing seasons. Assessments were made using at least 12 to 25 young

leaves per shoot (depending on available leaves per shoot) on four medium-height shoots of each vine in the middle portion of the plot. These leaves originated from the central portion of the vine shoots. Shoots with diseased leaves were marked with a plastic tag proximal to the shoot tip to ensure the same shoot and leaves were surveyed throughout the season. Downy mildew incidence was defined as the number of leaves with downy mildew symptoms divided by the total number of leaves evaluated. Downy mildew severity was assessed visually with the aid of a diagrammatic scale (Buffara et al. 2014).

Downy mildew disease progress curves were constructed and four epidemiological variables were compared: beginning of symptoms appearance (BSA), time to reach maximum disease incidence/severity (TRMDI and TRMDS), maximum disease incidence/severity ($I_{\max}$ and $S_{\max}$), and area under the incidence/severity disease progress curve (AUDPC and AUDSPC). Percent incidence data were transformed to arcsine multiplied by the square root of percentage to normalize the data (Jeager 2008). AUDPC was calculated as $\Sigma((Y_i + Y_{i+1})/2)(t_{i+1} - t_i)$, where: Y = disease intensity (incidence or severity), t = unit of time, and i = cumulative number of evaluations. This area represented the trapezoidal integration value of severity (Campbell and Madden 1990).

Downy mildew incidence and severity progression data were fit with three empirical models, monomolecular ($y = 1 -$

$(1-y_0)^{-rt}$), logistic ($y = 1/(1 + ((1/y_0) - 1)^{-rt})$), and Gompertz ($y = \exp(-(-\ln(y_0))^{-rt})$), where: y = incidence or severity (relative proportion from 0 to 1) over time t , y_0 = initial disease level and r = rate of disease increment for each empirical model. Since the monomolecular model is typically meant to fit monocyclic disease progression data, it was included in the current analysis as a form of negative control. Fit of each model to the data was evaluated based on the coefficient of determination (R^2) and residual error. Estimated parameters from each model were compared according to training systems and cultivars using a t test ($\alpha=0.05$) (Madden et al. 2007). Tests for significance were performed using an analysis of variance and post-hoc comparisons with the Tukey-Kramer HSD Test ($\alpha=0.05$). SAS software, version 9.1 (SAS Institute, Cary, NC, USA), was used for all statistical analyses.

Results

Average temperatures and rainfall were similar across the two growing seasons (Figs. 2 and 3). During the period of disease assessments from October to March of the 2012/2013 and 2013/2014 growing seasons, the maximum, average and minimum temperatures and rainfall were 24.4, 18.5, 13.9 °C and 874.2 mm and 25.4, 19.1, 14.1 °C and 863.1 mm, respectively (Figs. 2 and 3). The average of the maximum, mean and minimum daily temperatures and rainfall of these periods of evaluation were 24.9, 18.8, 14.0 °C and 868.6 mm, respectively (Figs. 2 and 3). A combination of frequent rain, mild temperature and sustained humidity, particularly in spring and summer, provided conditions for leaves to remain wet, thus enabling infections by *P. viticola* to establish in all cultivars.

Downy mildew incidence and severity remained low until the first week of January (beginning of ripening) for both cultivars and growing seasons in all training

systems. These corresponded to approximately 28 days after the first evaluation (DAFE) in the first growing season and 42 DAFE in the second growing season (Figs. 2 and 3).

Epidemiological variables were estimated for each training system according to cultivar and year (Tables 1 and 2). Among all training systems and cultivars, beginning of symptoms appearance (BSA) was 5 December, 2012 and 20 November, 2013, corresponding to the phenological stages (BBCH-scale) of fruit set in 2012 and full flowering in 2013. BSA during the 2012–2013 growing season started first at 28 DAFE when compared with the 2013–2014 growing season, probably because the rainfall increased and optimal environmental conditions for polycyclic disease development were reached. The time to reach maximum disease incidence (TRMDI) was roughly 15 days shorter for 2012/13 (approx. 50 days) as compared to 2013 (approx. 75 days). Similarly, time to reach maximum disease severity (TRMDS) was about 10 days shorter in 2012/2013 (approx. 84 vs. 95 days). Although there was no apparent trend according to cultivar or training system in 2013/2014, in the 2012/2013 season, cv ‘Cabernet Sauvignon’ reached maximum disease incidence about 5 days sooner and maximum severity about 6 days sooner than cv ‘Merlot’ (Tables 1 and 2).

The main variable that differentiated training systems was AUSDP and $S_{\max}$. Maximum disease severity showed a similar trend, where VSP yielded the lowest disease severity for both cultivars in both years (average $S_{\max}=25.1$) and SC was lowest in both cultivars in 2013/2014 (average $S_{\max}=59.2$) (Table 2). In all cases, disease severity variables ($S_{\max}$ and AUSDP) estimated for the VSP training system were significantly lower ($P<0.05$) than the SC training system. In all cases except one (cv ‘Merlot’ trained to GDC in 2012/2013), there was no significant difference between the cumulative disease

Fig. 2 Daily accumulated rainfall (mm), maximum and minimum temperature (°C), during growing season of 2012–2013, at Lages municipality, southern Brazil

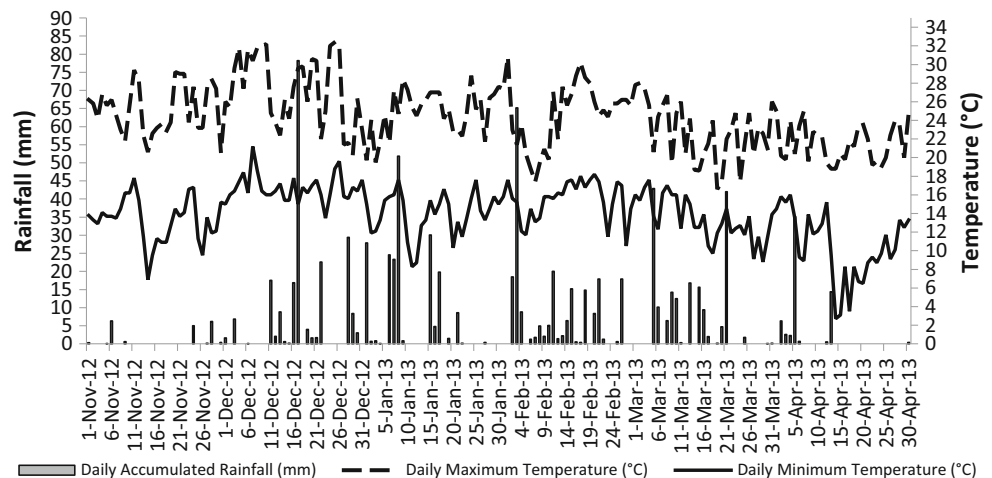
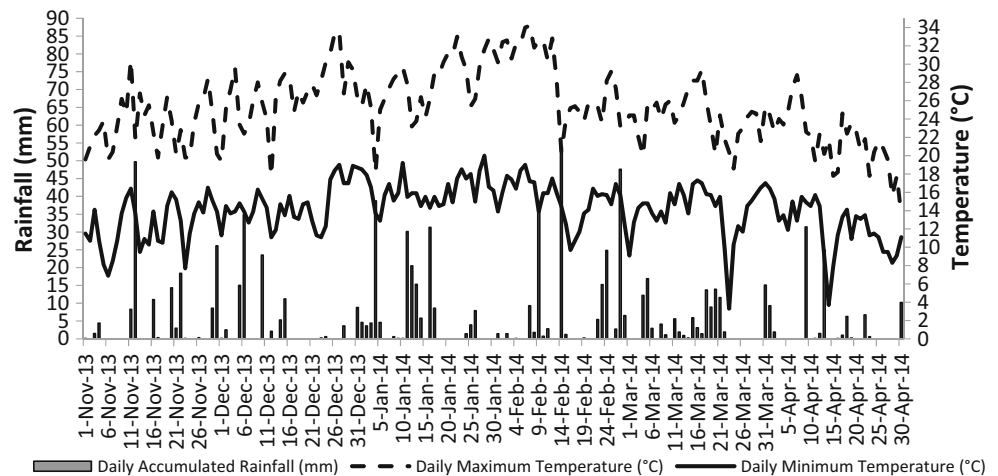


Fig. 3 Daily accumulated rainfall (mm), maximum and minimum temperature (°C), during growing season of 2013–2014, at Lages municipality, southern Brazil



severity under the GDC and Tendone training systems. Disease intensity represented by AUIDPC and AUSDPC was also significantly higher ($P < 0.05$) in the Tendone and GDC training system vines in both growing seasons (Tables 1 and 2). These differences were more pronounced when disease levels were relatively low, consequently, in the 2012–2013 growing season that had relatively high levels of downy mildew, the differences between the two Tendone and GDC training systems were less evident (Figs. 4 and 5).

In general, there was no significant difference in the estimated variables according to cultivar (Tables 1 and 2). For example, the average apparent infection rate was very similar within each year and cultivar (cv ‘Cabernet Sauvignon’ $r = 1.02$ and cv ‘Merlot’ $r = 1.03$ in 2012/2013; cv ‘Cabernet Sauvignon’ $r = 0.91$ and cv ‘Merlot’ $r = 0.97$ in 2013/2014) (Table 3) and there was no difference according to year, cultivar, or training system in BSA, which ranged from 28 to 42 days (Tables 1 and 2). Epidemiological models showed good fit ($P < 0.05$; Table 3) to the progress data of disease incidence and severity for each

Table 1 Epidemiological variables of downy mildew disease epidemics: beginning of symptom appearance (BSA), time to reach the maximum disease incidence (TRMDI) and severity (TRMDS), maximum value of disease incidence (I_{max}) and severity (S_{max}) and area under the disease incidence progress curve (AADIPC) and severity (AADSPC). Disease was evaluated in ‘Cabernet Sauvignon’ and ‘Merlot’ grape under tendone, simple curtain (SC), vertical shoot positioning (VSP), Geneva double curtain (GDC) during the 2012–2013 growing seasons at Lages municipality/SC, southern Brazil

Growing season 2012/2013					
Epidemiological variables	Tendone	SC	VSP	GDC	C.V. (%)
Cabernet Sauvignon					
BSA (days)	38.5 Aa ³	39.4 Aa	36.7 Aa	41.3 Aa	10.4
TRMDI (days)	46.9 Ab	47.6 Aa	47.6 Aa	47.6 Aa	7.1
TRMDS (days)	82.6 Aa	82.6 Aa	79.1 Ab	80.5 Aa	5.9
I_{max}	99.6 Aa	99.8 Aa	72.9 Ba	99.6 Aa	4.3
S_{max} ¹	90.5 Aa	93.5 Aa	22.3 Ba	96.9 Aa	4.3
AUIDPC ²	96.4 Aa	97.8 Aa	64.2 Ba	94.1 Aa	5.3
AUSDPC ²	2881.5Ab	1187.5 Bb	629.1 Ca	3126.4 Ab	11.9
Merlot					
BSA (days)	36.7 Aa	37.8 Aa	41.3 Aa	39.9 Aa	14.6
TRMDI (days)	58.1 Aa	53.9 Aa	49.9 Aa	49.0 Aa	5.7
TRMDS (days)	87.7 Aa	86.8 Aa	90.1 Aa	85.4 Aa	4.5
I_{max}	99.1 Aa	98.2 Aa	71.4 Ba	100.0 Aa	5.6
S_{max} ¹	89.2 Aa	77.1 Ab	25.5 Ba	82.7 Ab	2.7
AUIDPC ²	85.9 Aa	89.2 Aa	61.6 Ba	96.1 Aa	5.4
AUSDPC ²	4109.1 Aa	2169.5 Ba	492.0 Cb	4622.9 Aa	8.1

¹ Percentage of leaf area infected by a diagrammatic scale of Buffara *et al.* (2014)

² Area calculated by trapezoidal integration value according to Campbell and Madden (1990)

³ Means followed by the same capital letter in the line among each training systems and the lower case letter in the column among each variety are not significant different (Tukey-Kramer HSD test, $P > 0.05$)

Table 2 Epidemiological variables of downy mildew disease epidemics: beginning of symptom appearance (BSA), time to reach the maximum disease incidence (TRMDI) and severity (TRMDS), maximum value of disease incidence (I_{max}) and severity (S_{max}) and area under the disease incidence progress curve (AADIPC) and severity (AADSPC). Disease was evaluated in ‘Cabernet Sauvignon’ and ‘Merlot’ grape under tendone, simple curtain (SC), vertical shoot positioning (VSP), Geneva double curtain (GDC) during the 2013–2014 growing seasons at Lages municipality/SC, southern Brazil

Growing season 2013/2014					
Epidemiological variables	Tendone	SC	VSP	GDC	C.V. (%)
Cabernet Sauvignon					
BSA (days)	42.0 Aa ³	39.2 Aa	37.6 Aa	28.0 Ba	21.7
TRMDI (days)	70.7 Aa	82.6 Aa	73.5 Aa	78.8 Aa	8.7
TRMDS (days)	89.6 Aa	93.8 Aa	91.0 Aa	96.8 Aa	5.1
I_{max}	98.3 Aa	97.9 Aa	74.9 Ba	98.2 Aa	5.6
S_{max} ¹	78.2 Ba	53.4 Bb	27.5 Ca	96.4 Aa	36.8
AUIDPC ²	83.2 Aa	80.5 Aa	62.7 Ba	76.0 Aa	9.7
AUSDPC ²	2355.6 Ab	1280.1 Bb	459.6 Ca	2212.7 Ab	21.2
Merlot					
BSA (days)	35.2 Ab	40.6 Aa	26.6 Bb	33.8 Aa	24.2
TRMDI (days)	69.5 Ba	73.5 Aa	79.8 Aa	74.9 Aa	4.9
TRMDS (days)	95.2 Aa	97.3 Aa	96.6 Aa	95.7 Aa	2.3
I_{max}	99.6 Aa	97.8 Aa	72.7 Ba	100.0 Aa	4.7
S_{max} ¹	87.9 Aa	65.0 Ba	25.1 Ca	86.7 Aa	21.3
AUIDPC ²	77.9 Aa	74.0 Aa	65.7 Ba	76.7 Aa	6.7
AUSDPC ²	3919.6 Aa	1859.0 Ca	318.6 Db	2853.6 Ba	29.1

¹ Percentage of leaf area infected by a diagrammatic scale of Buffara *et al.* (2014)

² Area calculated by trapezoidal integration value according to Campbell and Madden (1990)

³ Means followed by the same capital letter in the line among each training systems and the lower case letter in the column among each variety are not significant different (Tukey-Kramer HSD test, $P \leq 0.05$)

season-by-year combination. Gompertz model showed the best fit to the data (Table 3). In general, the infection rate estimated

by the Gompertz model was higher in 2012/2013 (average $r=1.025$) than in 2013/2014 (average $r=0.940$) (Table 3).

Fig. 4 Areas under disease progress curve of downy mildew for Tendone, Simple Curtain (SC), Vertical Shoot Positioning (VSP) and Geneva Double Curtain (GDC) in cv ‘Cabernet Sauvignon’ during growing seasons of 2012–2013 and 2013–2014 in Lages, SC municipality, southern Brazil. Area under severity disease progress curve (AUSDPC) growing seasons of 2012–2013 (a); Area under incidence disease progress curve (AUIDPC) growing seasons of 2012–2013 (b); AUSDPC growing season of 2013–2014 (c) and AUIDPC growing season of 2013–2014 (d)

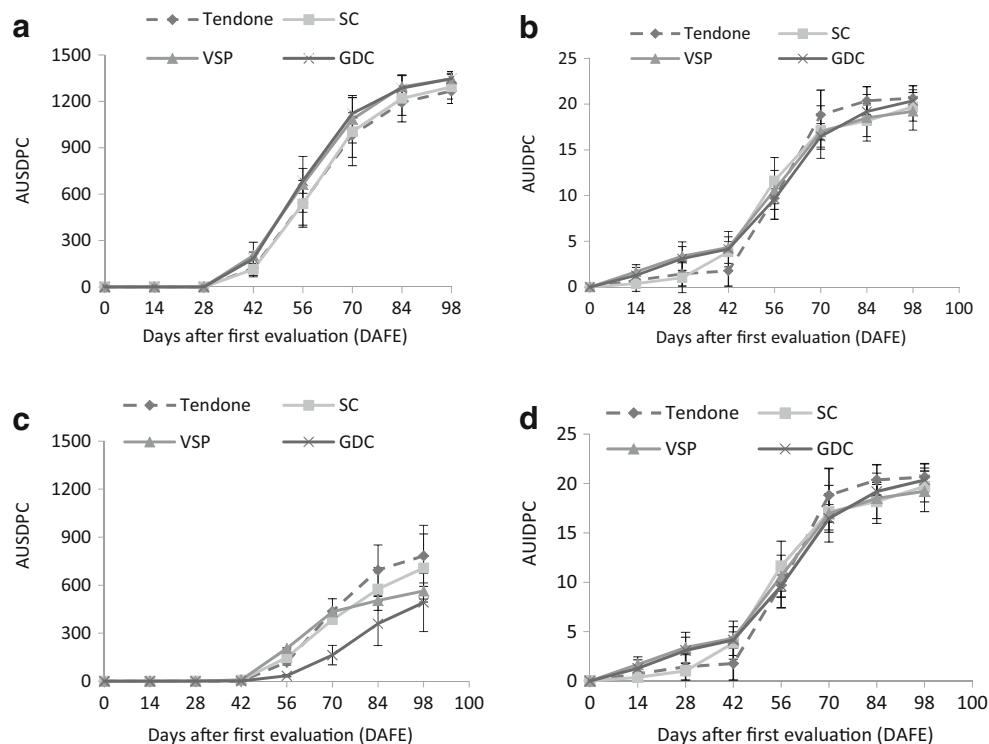
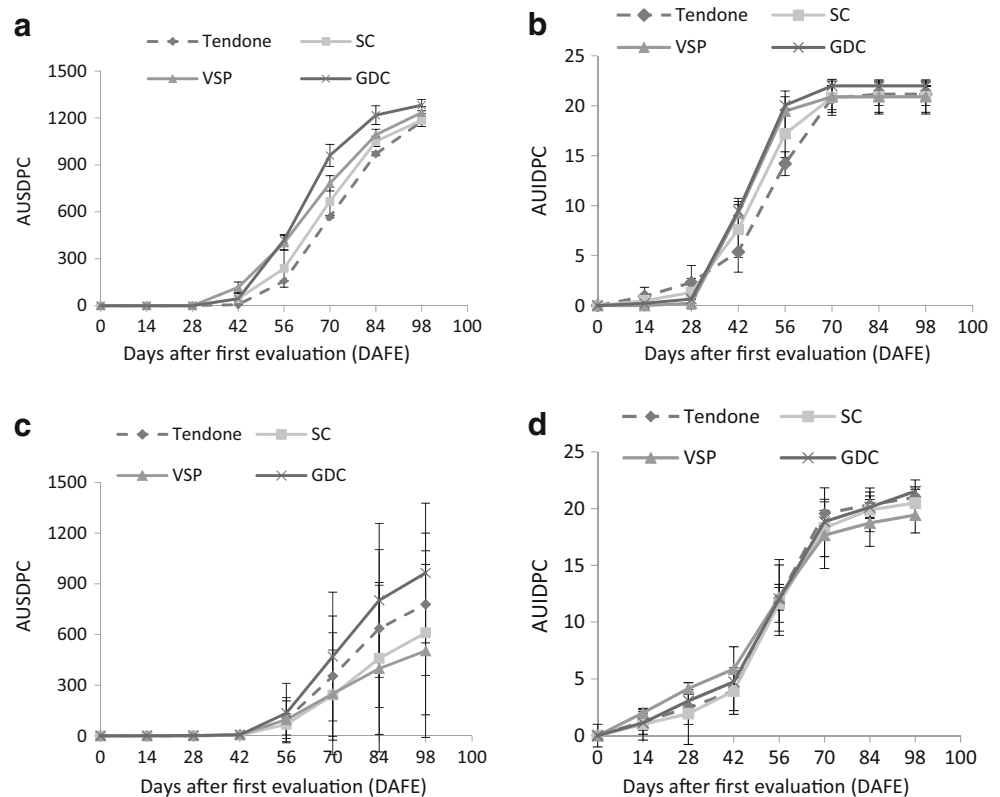


Fig. 5 Areas under disease progress curve of downy mildew for Tendone, Simple Curtain (SC), Vertical Shoot Positioning (VSP) and Geneva Double Curtain (GDC) in cv ‘Merlot’ during growing seasons of 2012–2013 and 2013–2014 in Lages, SC municipality, southern Brazil. Area under severity disease progress curve (AUSDPC) growing seasons of 2012–2013 (a); Area under incidence disease progress curve (AUIDPC) growing seasons of 2012–2013 (b); AUSDPC growing season of 2013–2014 (c) and AUIDPC growing season of 2013–2014 (d)



Discussion

Results of this study were consistent with previous findings reporting differences among cultivars and training systems with respect to grape downy mildew intensity (De Bem et al. 2015). In our studied region, downy mildew inoculum was never a limiting factor given the historical occurrence of this disease. The high rainfall in the months prior to the beginning of symptoms appearance till harvest in the first season likely explained the increasing levels of downy mildew disease severity. The percent of infected leaves on both cultivars were generally high, indicating that both cultivars are susceptible to downy mildew. In general, downy mildew disease incidence and severity were higher and average number of days to reach I_{max} and S_{max} were lower in the 2012/13 season, which was characterized as having more frequent and regular precipitation as compared to 2013/2014. Thus, it is likely that the increase of rainfall during the 2012/2013 growing season, especially between December to March (end of flowering to harvest) in both growing seasons was a major contributing factor for overall increase in downy mildew severity. Indeed, precipitation as a driving force for increased disease incidence and severity is supported by the model proposed by Lalancette et al. (1988), which gives greater weight to leaf wetness duration than to average temperatures for estimating disease severity. Thus, duration of leaf wetness can be viewed as the factor that allows infection to take place, whereas temperature

determines the rate and extent of infection (Madden et al. 2000; Gindro et al. 2006).

Comparison of epidemiological variables estimated under each training system within each cultivar indicated cumulative measures of disease incidence and disease severity to be most discriminating. These two variables indicated VSP achieved lowest disease across both years and all cultivars, where AUSDPC was significantly lower than all other training systems for both cultivars in both years. In the SC training system comparatively low disease levels were observed, resulting in AUSDPC significantly lower than both GDC and Tendone training systems (with the exception of ‘Merlot’ trained to GDC in 2013). Thus, it can be concluded that shoot positioning and subsequent canopy alteration serves to limit (VSP and SC) or enhance (Tendone and GDC) the severity of disease development within both cultivars.

VSP is a common and widely used trellis system worldwide and ideal for low-vigor vines. Shoots are trained upward in a narrow curtain with the fruiting zone below. In comparison, GDC divides the canopy horizontally, with the goal to reduce the canopy density, allowing more sunlight to reach the fruit zone. With this system, grapevines are trained from the trunk to bilateral cordons and pruned to retain short canes (1.80 to 2.40 m) originating preferably in the lower 180° of the cordon. The GDC system is similar, but requires a three-wire trellis, using two horizontal cordon-support wires and a

Table 3 Epidemiological model equations and adjusted coefficient of determination (R^2) fit with Monomolecular, Logistic and Gompertz models to downy mildew severity observed in Tendone, Simple Curtain (SC), Vertical Shoot Positioning (VSP) and Geneva Double Curtain (GDC) at the ‘Cabernet Sauvignon’ and ‘Merlot’ grape during 2012–2013 and 2013–2014 growing seasons in Lages, SC municipality, southern Brazil

Cultivar	Epidemiological models					
	Monomolecular ^a		Logistic ^b		Gompertz ^c	
Growing season 2012/2013						
Cabernet Sauvignon	Equation	<i>R</i> ²	Equation	<i>R</i> ²	Equation	<i>R</i> ²
Tendone	y = 0.772x + 0.113	0.812	y = 0.958x - 0.082	0.833	y = 1.0222x + 0.014	0.981
SC	y = 0.710x + 0.152	0.778	y = 0.950x - 0.087	0.821	y = 1.022x + 0.013	0.987
VSP	y = 0.646x + 0.207	0.769	y = 0.969x - 0.103	0.847	y = 1.015x + 0.010	0.996
GDC	y = 0.647x + 0.209	0.759	y = 0.933x - 0.097	0.796	y = 1.018x + 0.014	0.991
Merlot	Equation	<i>R</i> ²	Equation	<i>R</i> ²	Equation	<i>R</i> ²
Tendone	y = 0.649x + 0.131	0.694	y = 0.988x - 0.043	0.929	y = 1.055x - 0.025	0.993
SC	y = 0.708x + 0.116	0.745	y = 1.014x - 0.064	0.901	y = 1.032x - 0.006	0.989
VSP	y = 0.693x + 0.145	0.795	y = 1.033x - 0.101	0.881	y = 0.991x + 0.021	0.995
GDC	y = 0.692x + 0.150	0.730	y = 0.921x - 0.070	0.807	y = 1.042x + 0.0003	0.982
Growing season 2013/2014						
Cabernet Sauvignon	Equation	<i>R</i> ²	Equation	<i>R</i> ²	Equation	<i>R</i> ²
Tendone	y = 0.925x + 0.0133	0.763	y = 1.191x - 0.053	0.888	y = 0.910x + 0.038	0.902
SC	y = 0.917x + 0.012	0.752	y = 1.423x - 0.054	0.808	y = 0.921x + 0.021	0.954
VSP	y = 0.978x + 0.002	0.727	y = 1.525x - 0.046	0.823	y = 0.802x + 0.028	0.985
GDC	y = 0.921x + 0.009	0.718	y = 1.253x - 0.030	0.800	y = 0.992x + 0.010	0.953
Merlot	Equation	<i>R</i> ^{*2}	Equation	<i>R</i> ^{*2}	Equation	<i>R</i> ^{*2}
Tendone	y = 0.875x + 0.027	0.721	y = 1.159x - 0.048	0.815	y = 0.986x + 0.017	0.957
SC	y = 0.902x + 0.015	0.736	y = 1.277x - 0.044	0.834	y = 0.974x + 0.013	0.968
VSP	y = 0.946x + 0.008	0.794	y = 1.374x - 0.047	0.884	y = 0.895x + 0.023	0.937
GDC	y = 0.815x + 0.053	0.790	y = 1.029x - 0.042	0.876	y = 1.040x - 0.0003	0.978

^a Monomolecular $y = 1 - (1 - y_0) \exp(-rt)$ ^b Logistic $y = 1 / (1 + ((1/y_0) - 1) \exp(-rt))$ ^c Gompertz $y = \exp(-(-\ln(y_0)) \exp(-rt))$, where y = severity in proportion of 0 to 1 in time t and, y_0 = initial level of disease and r = disease increment rate for each empirical model

lower single trunk support wire (Ahmedullah 1996). The SC system is the least expensive of the different trellising systems. It is adapted to mechanical harvesting and typically used for Concord (*V. labrusca* L.) grapes. Grapes are cordon trained along this wire in both directions (Ahmedullah 1996). Finally, the Tendone system is designed to create a large horizontal canopy where primary shoots are tied to a wire grid that is parallel to the soil at 1.8 m height, while the lateral shoots grow upward and the clusters hang below the trellis (Miele et al. 2009).

The Tendone and GDC systems are known to increase bud fruitfulness, influencing the amount of fruit that is produced, regardless of cultivar. This leads to production of a dense vegetative canopy, which produces favorable microclimatic conditions for downy mildew disease development (Gindro et al. 2006). Similarly, although some form of hedging or shoot trimming is necessary for each of these systems, under

moderate- to high-vigor conditions, the effects of this practice on canopy microclimate may be temporary due to compensating lateral shoot growth. However, shoot positioning under the VSP training system allow for maintaining canopy shape and foliage separation when there is narrow row spacing.

The maintenance of an open canopy under the VSP and SC training systems may be one of the most significant factors limiting overall downy mildew disease (AUIDPC and AUSDPCC) under these systems. Creating and maintaining an open canopy improves light penetration to the canopy interior, particularly in vigorous, vertically divided vineyards where the row middle or area between the fruiting zones becomes shaded following fruit set. Advantages of the VSP system are that it can make cultural practices easier, such as leaf removal, shoot removal and cluster thinning, while also providing for more efficient spray coverage (Calonnec et al. 2013). For horizontally divided systems, shoot positioning is normally

performed once per year near bloom (Dokoozlian and Kliewer 1995). While some shading is beneficial, especially in very hot and sunny climates, to prevent heat stress, excessive amounts of shading can have positive downy mildew development and negative impact on grape development.

The VSP and SC training systems showed the lowest cumulative downy mildew development in both cvs ‘Cabernet Sauvignon’ and ‘Merlot’. It is particularly suited to the natural upward growth of *V. vinifera* vines. VSP also has the advantage of being compatible with vineyard mechanization. However, a drawback to the VSP system is that it may be more difficult to manage with high vigor cultivars or on fertile sites that produce large vines. In this case, leaf removal can help to keep the canopy from becoming too dense and to reduce disease. However, in conditions with high amounts of sunlight and high heat, managers must take precaution not to remove leaves on the afternoon side of the canopy to prevent sunburn. In those cases, canopy management on a VSP is practiced only on the morning side of the canopy (Domine 2008).

An appropriate empirical model, which describes the disease progress curve, can be selected from the disease temporal analysis. This selection aimed to estimate variables that can be used to compare epidemics (Madden et al. 2007). Different epidemiological models were fit to the data obtained for each cultivar under each training system. For both cultivars and across both seasons, the Gompertz model fit the data best. According to this model, the rate of disease progression is proportional to pre-existing inoculum (initial inoculum) and infection rate and is appropriate for polycyclic diseases (Madden et al. 2007). Compared to the logistic model, the Gompertz model estimated the maximum rate of disease development (dy/dt) earlier, which may explain better fit of this model.

Canopy management has become an integral component of an integrated disease management strategy for grapevine. Our results support the predominant practice of microclimate management by canopy alteration using a single high-wire training system provided that an upright shoot growth pattern. The adoption of a coiled support wire and practicing timely shoot trimming are crucial factors in promoting such a growth pattern even in the most recalcitrant varieties like Trebbiano and Pinot types (Louarn et al. 2008; Pugliese et al. 2011). Collectively, the results of this study suggest VSP training system should be recommended for grapevine production to reduce downy mildew in highlands regions of southern Brazil.

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References

- Ahmedullah M (1996) Training and trellising grapes for production in Washington State. Washington State University Extension Publication # EB0637. Pullman
- Alonso-Villaverde V, Boso S, Santiago JL, Gago P, Rodriguez-García MI, Martínez MC (2011) Leaf thickness and structure of *Vitis vinifera* cv. Albariño clones and its possible relation with downy mildew (*Plasmopara viticola*) infection. Int J Vine Wine Sci 45: 161–169
- Austin CN, Wilcox WF (2011) Effects of fruit-zone leaf removal, training systems, and irrigation on the development of grapevine powdery mildew. Am J Enol Vitic 62:193–198
- Boso S, Alonso-Villaverde A, Gago P, Santiago JL, Martínez MC (2011) Susceptibility of 44 grapevine (*Vitis vinifera* L.) varieties to downy mildew in the field. Aust J Grape Wine Res 17: 394–400
- BRASIL (2013) Ministério da Agricultura, do Abastecimento e da Reforma Agrária. Uva para exportação: aspectos técnicos da produção. Brasília: EMBRAPA-SPI. Série Publicações 359 Técnicas FRUPEX n° 25
- Buffara CRC, Angelotti F, Vieira FA, Bogo A, Tessmann DJ, de Bem BP (2014) Elaboration and validation of a diagrammatic scale to assess downy mildew severity in grapevine. Cienc Rural 44:1384–1391
- Calonnec A, Burie JB, Langlais M, Guyader S, Saint-Jean S, Sache I, Tivoli B (2013) Impacts of plant growth and architecture on pathogen processes and their consequences for epidemic behavior. Eur J Plant Pathol 135:479–497
- Campbell CL, Madden LV (1990) Introduction to plant disease epidemiology. Wiley Press, New York
- Chavarria G, Santos HP, Sônego OR, Marodin GS, Bergamaschi H, Cardoso LS (2007) Incidence of diseases and needs of control in overhead covered grapes. Rev Bras Frutic 29:477–482
- De Bem BP, Bogo A, Everhart S, Casa RT, Gonçalves MJ, Marcon Filho JL, da Cunha IC (2015) Effect of Y-trellis and vertical shoot positioning training systems on downy mildew and botrytis bunch rot of grape in highlands of southern Brazil. Sci Hortic 185:162–166
- Dokoozlian NK, Kliewer WM (1995) The light environment within grapevine canopies. I Description and seasonal changes during fruit development. Am J Enol Vitic 46:209–218
- Domine A (2008) Wine. Ullmann Publishing Press, New York
- Epagri (2015) Centro de Informações de Recursos Ambientais e de Hidrometeorologia de Santa Catarina. Available: http://ciram.epagri.sc.gov.br/index.php?option=com_content&view=article&id=98&Itemid=198. Accessed 13 April 2015
- Gindro K, Spring JL, Pezet R, Richter JO (2006) Histological and biochemical criteria for objective and early selection of grapevine cultivars resistant to *Plasmopara viticola*. Vitis 45:191–196
- Giorio P, Nuzzo V (2012) Leaf area, light environment, and gas exchange in Montepulciano grapevines trained to Tendone trellising system. Plant Biosyst 146:322–333
- Howel GS, Miller DP, Edson CE, Striegler RK (1991) Influence of training system and pruning severity on yield, vine size, and fruit composition of Vignoles grapevines. Am J Enol Vitic 42: 191–198
- Jaeger TF (2008) Categorical data analysis: away from ANOVAs (transformation or not) and towards Logit Mixed Models. J Mem Lang 59:434–446
- Kennelly MM, Gadoury DM, Wilcox HF, Magarey PA, Seem RC (2005) Seasonal development of ontogenic resistance to downy mildew in grape berries and rachises. Phytopathology 95:1445–1452

- Lalancette N, Ellis MA, Madden LV (1988) Development of an infection efficiency model for *Plasmopara viticola* on American grape based on temperature and of leaf wetness. *Phytopathology* 78:794–800
- Louarn G, Lecoeur J, Lebon E (2008) A three-dimensional statistical reconstruction model of grapevine (*Vitis vinifera* L.) simulating canopy structure variability within and between cultivar/training system pairs. *Ann Bot* 101:1167–1184
- Madden LV, Ellis MA, Lanlancette N, Wilson LL (2000) Evaluation of a disease warning system for downy mildew of grapes. *Plant Dis* 84: 549–554
- Madden LV, Hughes G, Van den Bosch F (2007) The study of plant disease epidemics. American Phytopathological Society, St Paul
- Miele A, Rizzon LA, Mandelli F (2009) Grapevine canopy management effects on Merlot wine composition. *Pesq Agrop Brasileira* 44:463–470
- Peel MC, Finlayson BL, McMahon TA (2007) Updated world map of the Koppen-Geiger climate classification. *Hydrol Earth Syst Sci* 11: 1633–1644
- Pugliese M, Gullino ML, Garibaldi A (2011) Effect of climate change on infection of grapevine by downy and powdery mildew under controlled environment. *Commun Agric Appl Biol Sci* 76:579–582
- Reynolds AG, Wardle DA, Cliff MA, King M (2004) Impact of training system and vine spacing on vine performance, berry composition, and wine sensory attributes of Seyval and Chancellor. *Am J Enol Vitic* 55:84–954
- Yu Y, Zhang Y, Lu J (2012) The mode of host resistance to *Plasmopara viticola* infection of grapevines. *Phytopathology* 102:1094–1101
- Zahavi T, Reuveni M, Scheglov D, Lavee S (2001) Effect of grapevine training systems on development of powdery mildew. *Eur J Plant Pathol* 107:495–501